

Using common Maclaurin's series

Key Points

You must be able to find the list of Maclaurin's series in your formula booklet

Know that you can substitute x for any other function $f(x)$ in any known Maclaurin's series (including any limit placed on x)

e.g.

$$\begin{aligned}\cos(\sin x) &= 1 - \frac{(\sin x)^2}{2} + \frac{(\sin x)^4}{24} + \dots \\ &= 1 - \frac{\left(x - \frac{x^3}{6}\right)^2}{2} + \frac{\left(x - \frac{x^3}{6}\right)^4}{24} + \dots = \dots\end{aligned}$$

Basic Skills

Q1. State the Maclaurin's series for the following

- a) $\sin x$
- b) $\cos x$
- c) e^x
- d) $\ln(1+x)$

Q2. State which Maclaurin's series in the formula book have the a limit on the value of x

Q3. By replacing x with $2x$ find a Maclaurin's series for e^{2x}

Q4. By replacing x with $\frac{x}{2}$ find a Maclaurin's series for $\sin\left(\frac{x}{2}\right)$

Competency Skills

Q5. Find the first three non-zero terms of the Maclaurin's series for $e^{\sin x}$

Q6. a) Find the Maclaurin's series for $\cos(x^2)$

b) Find the Maclaurin's series for $\sin(x^2)$

c) Find the Maclaurin's series for $\tan(x^2)$ up to the x^6 term

Q7. a) Find the first three non-zero terms of the Maclaurin's series for $f(x) = \ln(1 + \sin x)$

b) State the domain for which this approximation can be used

c) Find an approximation for $f\left(\frac{\pi}{3}\right)$ and state the percentage error

Advanced Skills

Q8. Why do we not have a standard Maclaurin's series for $\ln x$

Q9. Find the Maclaurin's series for $f(x) = \ln(1 + e^x)$ up to the x^2 term and state the domain for which this series can be used

Q10. Find the Maclaurin's series for $f(x) = \ln(e^{-3x} + e^{3x})$ up to the x^2 term

Q11. By using Maclaurin's series' find the limit of $\frac{\sin(x^2) - x^2 + \frac{1}{6}x^6}{x^{10}(x^5 + 5)}$ as $x \rightarrow 0$

Q12. Given that the first three terms of the Maclaurin's series for $e^{3x} \cos 2x$ and $(1 + ax)^n$ are equal find the constants a and n

Extension

Q12. Expand your formula book to contain other useful functions by calculating the Maclaurin's series for common functions (try hyperbolic trig functions e.g. $\sinh x = \frac{e^x - e^{-x}}{2}$)